



CECILIA PAYNE- GAPOSCHKIN

British-American astronomer
1900–1979

Our nearest star—the sun—is an unimaginable 93 million miles (150 million km) away. Today’s spacecraft cannot get much closer than 7 million miles (11 million km) to it without melting. Yet, we’ve known for almost 100 years what this fiery star is made of. Cecilia Payne-Gaposchkin was the star scientist who solved this puzzle.

Cecilia figured out the sun’s secrets from Earth, using special photographs of stars, called spectra. These photographs split the white light from a star into the rainbow of different colors of light that it’s made from. Cecilia found a new way to read these spectra.



As a child, Cecilia spent hours in the library exploring her favorite subjects: science and mathematics.

These dark lines, known as Fraunhofer lines, show different colors that are missing from this spectrum. The missing colors can be used to figure out which elements are present in the spectrum.

This missing color has been absorbed by hydrogen.

This missing color has been absorbed by iron.